



# Matrix Theory

## 12 Orthogonality

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July-Nov 2026

# Previously



భారతీయ సాంకేతిక విజ్ఞాన సంస్థ హైదరాబాద్  
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- Vector Spaces

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- Vector Spaces
  - Linear Independence
  - Spanning
  - Basis, and Dimension



- Set of independent vectors that spans a space



- Set of independent vectors that spans a space
- Geometrically: set of coordinate axes



- Set of independent vectors that spans a space
- Geometrically: set of coordinate axes
- Whenever we imagine a space like  $\mathbb{R}^2$ , the coordinate axes are **perpendicular!**



When choosing a basis, we tend to use an **orthogonal** one.



Geometric constructions  $\xrightarrow{\text{Basis}}$  Algebraic calculations



Geometric constructions  $\xrightarrow{\text{Basis}}$  Algebraic calculations

- Orthogonal basis makes this simple



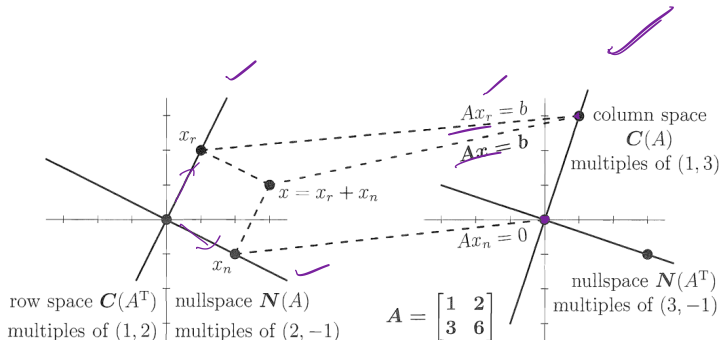
Geometric constructions  $\xrightarrow{\text{Basis}}$  Algebraic calculations

- Orthogonal basis makes this simple
- Making them unit vectors (**Orthonormal basis**) is favorable.

# We have seen perpendicularity

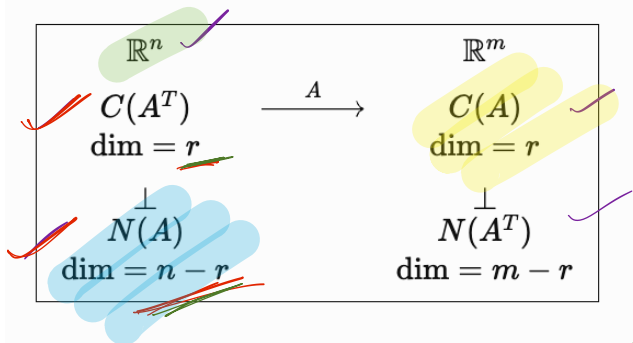


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Subspaces can be perpendicular!

# We have seen perpendicularity



$r$   
 $n-r$

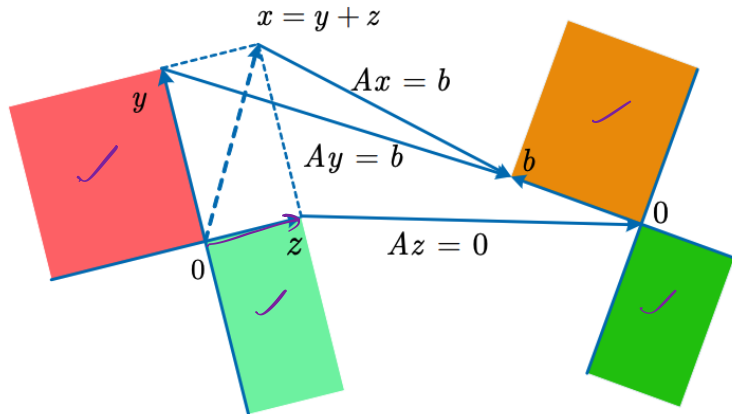
$$\mathbb{R}^n = C(A^T) \oplus N(A)$$

$$\mathbb{R}^m = C(A) \oplus N(A^T)$$

Orthogonal  
Complements

Subspaces can be perpendicular!

# We have seen perpendicularity



Fundamental subspaces meet at right angles!

# Hinge question



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- Do you think the floor and a wall of our classroom form orthogonal subspaces?

NO. They have a line (1D subspace) in common.  
Those vectors are not  $\perp$  to themselves.

# For an Orthonormal basis, we need



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- ① Length of vectors  $\|x\|$

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- ① Length of vectors  $\|x\|$
- ② Test of perpendicularity

# For an Orthonormal basis, we need



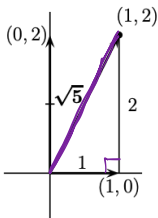
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- ① Length of vectors  $\|x\|$
- ② Test of perpendicularity
- ③ Independent vectors  $\overset{?}{\rightarrow}$  Perpendicular vectors

# Length of a vector ( $\|x\|$ )



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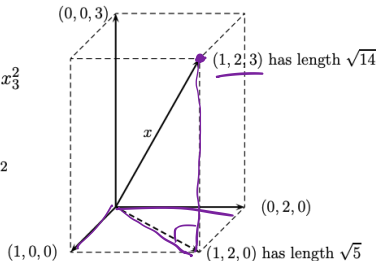


(a)

$$\|x\|^2 = x_1^2 + x_2^2 + x_3^2$$

$$5 = 1^2 + 2^2$$

$$14 = 1^2 + 2^2 + 3^2$$



(b)

Lengths of vectors in 2D and 3D using Pythagoras

# Length of a vector ( $\|x\|$ )



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- Extension to  $n$  dimensions: apply Pythagoras  $(n - 1)$  times

# Length of a vector ( $\|x\|$ )



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- Extension to  $n$  dimensions: apply Pythagoras  $(n - 1)$  times
- $\|x\|^2 = x_1^2 + x_2^2 + \dots + x_n^2$

# Length of a vector ( $\|x\|$ )



- Extension to  $n$  dimensions: apply Pythagoras  $(n - 1)$  times
- $\|x\|^2 = x_1^2 + x_2^2 + \dots + x_n^2$  ✓
- Sum of squares matches  $x^T x$  ✓

# Length of a vector ( $\|x\|$ )



- Extension to  $n$  dimensions: apply Pythagoras  $(n - 1)$  times
- $\|x\|^2 = x_1^2 + x_2^2 + \dots + x_n^2$
- Sum of squares matches  $x^T x$
- Length of a vector  $x = \|x\| = \sqrt{x^T x}$  ✓

# Determining perpendicularity

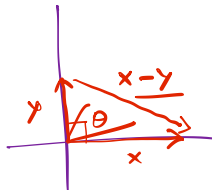


- Form a triangle and apply Pythagoras

$$\|x - y\|^2 = \|x\|^2 + \|y\|^2$$

$$-2(x_1y_1 + x_2y_2 + \dots + x_ny_n) = 0$$

$$\underline{x^T y = 0}$$



$$x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

$$\underline{\|x\|^2 + \|y\|^2 \geq \|x - y\|^2}$$

# Determining perpendicularity



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The inner product  $x^T y$  is zero iff  $x$  and  $y$  are orthogonal vectors. If  $x^T y > 0$ , their angle is acute. If  $x^T y < 0$ , their angle is obtuse.

# Some Notes



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- Length squares is inner product of  $x$  with itself:  $\|x\|^2 = x^T x$

# Some Notes



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- Length squares is inner product of  $x$  with itself:  $\|x\|^2 = x^T x$
- Only vector with length zero is 'zero vector'

# Some Notes



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- Length squares is inner product of  $x$  with itself:  $\|x\|^2 = x^T x$
- Only vector with length zero is 'zero vector'
- Only vector perpendicular to itself is 'zero vector'

# Some Notes



- Length squares is inner product of  $x$  with itself:  $\|x\|^2 = x^T x$
- Only vector with length zero is 'zero vector'
- Only vector perpendicular to itself is 'zero vector'
- 'Zero vector' is perpendicular to every vector in the space

# Mutually orthogonal $\rightarrow$ Linear Independence



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$$v_i \perp v_j \quad \forall i \neq j$$

If nonzero vectors  $\{v_1, v_2, \dots, v_k\}$  are mutually orthogonal (every vector is orthogonal to every other), then those vectors are linearly independent.

Consider  $c_i v_i = 0$ . Take dot product with a  $v_j$  on both sides

$$c_j \|v_j\|^2 = 0$$

$$\Rightarrow c_j = 0 \quad \forall j \quad \Rightarrow \text{L.I.}$$

# Curiosity question



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- Why not call perpendicular, why call Orthogonal?

# Curiosity question



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- Why not call perpendicular, why call Orthogonal?
- Generalizes the idea of a 90-degree angle to abstract spaces and higher dimensions

# Orthogonal subspaces



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Two subspaces **V** and **W** of the same space  $\mathbb{R}^n$  are orthogonal if every vector  $v$  in **V** is orthogonal to every vector  $w$  in **W**:  $v^T w = 0 \quad \forall v, w$

# Example: Orthogonal subspaces



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$\mathbb{R}^4$

- $V = \text{Span}\{(1, 0, 0, 0), (1, 1, 0, 0)\}$   $2D$  ✓
- $W = \text{Span}\{(0, 0, 4, 5)\}$   $1D$  ✓

$$z = c(0, 0, 5, -4) \quad 1D \quad \checkmark$$

$$c_1 \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$

$$d_1 \begin{bmatrix} 0 \\ 0 \\ 4 \\ 5 \end{bmatrix}$$

# Example: Orthogonal subspaces



- $V = \text{Span}\{ (1, 0, 0, 0), (1, 1, 0, 0) \}$  ✓
- $W = \text{Span}\{ (0, 0, 4, 5) \}$  ✓
- $L = \text{Span}\{ (0, 0, 5, -4) \}$  ✓

# Fundamental theorem of Orthogonality



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$$A_{m \times n}$$

The row space is orthogonal to null space (in  $\mathbb{R}^n$ ). The column space is orthogonal to the left null space (in  $\mathbb{R}^m$ ).

$$Ax = 0$$

$$\begin{bmatrix} \text{---} \\ \text{---} \\ \vdots \\ \text{---} \end{bmatrix} \quad \begin{matrix} | \\ | \\ | \\ | \end{matrix}$$

$m$

# Example: Orthogonal subspaces vs. Orthogonal complements



$\mathbb{R}^3$

- $U = \text{Span}\{(1, 0, 0)\}$  ✓ 1D
- $V = \text{Span}\{(0, 1, 1)\}$  ✓ 1D
- $W = \text{Span}\{(0, 1, 0), (0, 0, 1)\}$  ✓ 2D

$U, V$  are not  
ortho. Compl

$U \perp V$

$U \perp W$

$V \not\perp W$

$U, W$  are  
ortho. Compl



- Orthogonal complements

# Next



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- Orthogonal complements
- Part II of the fundamental theorem of orthogonality